Saving Time Without Losing Accuracy: Compress Then Explain on Temporal Data

An empirical study using the TabReD benchmark and CTE.

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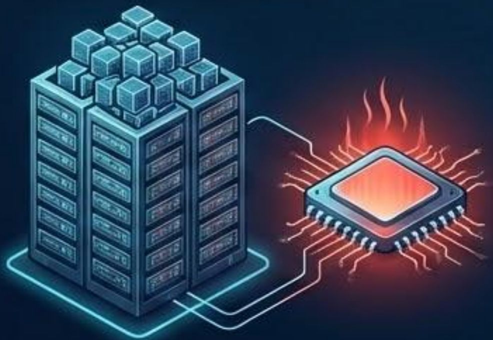
Problem

Computational Impracticality



Calculating exact machine learning explanations (like SHAP or SAGE) on massive datasets is computationally impractical.

Industry Workaround: Random Sampling

The standard industry workaround is random sampling (i.i.d.) from background data.

The Catch: Garbage In, Garbage Out

The Catch: "Garbage sample in, garbage explanation out". Random sampling introduces significant approximation errors and can even change feature importance rankings

Factorial 1 to 15

1! = 1	6! = 720	11! = 39,916,800
2! = 2	7! = 5040	12! = 479,001,600
3! = 6	8! = 40320	13! = 6,227,020,800
4! = 24	9! = 362880	14! = 87,178,291,200
5! = 120	10! = 3628800	15! = 1,307,674,368,000

Literature Review – Explainability

SHAP

Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions - foundational paper for background data samples estimations

SAGE

Ian Covert, Scott M Lundberg, and Su-In Lee. Understanding global feature contributions with additive importance measures - key global feature-importance method

Fooling SHAP

Gabriel Laberge, Ulrich Aivodji, Satoshi Hara, and Foutse Khomh Mario Marchand. Fooling SHAP with Stealthily Biased Sampling - showcases potential issues and concerns with standard i.i.d. sampling approach

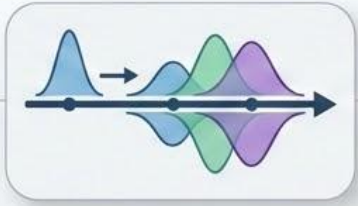
Literature Review - Kernel Thinning

- Raaz Dwivedi and Lester Mackey. **Kernel thinning** - main statistical motivation based improved distribution compression
- Raaz Dwivedi and Lester Mackey. **Generalized kernel thinning** - a generalized version of KT, foundational to Compress Then Explain approach
- Abhishek Shetty, Raaz Dwivedi, and Lester Mackey. **Distribution compression in near-linear time (COMPRESS++)** - efficient KT algorithm implementation

Main Research Questions

RQ1: Efficiency vs. Accuracy Trade-off

To what extent does the CTE paradigm reduce the computational overhead of generating SHAP explanations on massive TabReD datasets while maintaining high fidelity compared to standard i.i.d. sampling?



RQ2: Explanations Over Time (Concept Drift)

How do feature importances change chronologically in temporal datasets? Can a compressed background dataset accurately approximate explanations as the underlying data distribution shifts over time?

RQ3: Train vs. Validation Discrepancy

Does the model rely on fundamentally different features when evaluating unseen validation data compared to its training data, and how effectively does CTE highlight this?

Experimental Setup & Methodology

Our Pipeline:

- **Dataset:** Utilizing a temporal tabular dataset from TabReD.
- **Model:** Training a baseline model.
- **Explainer:** Implementing the Compress Then Explain (CTE) plugin for SHAP.

Evaluation:

- **Core Comparison:** Standard sampling vs. CTE.
- **Efficiency & Accuracy:** Trade-off between sample size, calculation time and SHAP approximation error.
- **Temporal Dynamics:** Tracking feature importance across time.
- **Generalization:** SHAP differences between Training and Validation datasets.

EXPECTED DELIVERABLES

Accelerating accurate SHAP estimations on temporal data while actively detecting model degradation and overfitting.

- Fully trained model on TabReD
- Performance benchmark (CTE vs. Standard SHAP)
- Temporal Drift visualizations
- Feature Importance stability plots (Training vs. Validation dataset)

FULLY TRAINED MODEL ON TABRED

PERFORMANCE BENCHMARK (CTE VS. STANDARD SHAP)

TEMPORAL DRIFT VISUALIZATIONS

FEATURE IMPORTANCE STABILITY PLOTS (TRAINING VS. VALIDATION DATASET)

ACCELERATING ACCURATE SHAP ESTIMATIONS ON TEMPORAL DATA WHILE ACTIVELY DETECTING MODEL DEGRADATION AND OVERFITTING.

References

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- Hubert Baniecki Giuseppe Casalicchio, Bernd Bischl, Przemyslaw Biecek (2025) Efficient and Accurate Explanation Estimation with Distribution Compression https://openreview.net/pdf?id=LiUfN9h0Lx

Thank You!

Questions & Feedback.